Remote Infant Studies of Early Learning (RISE): Expansion of an online battery for infant cognitive and language development

Ethical Statement: The research conducted here was in compliance with the American Psychological Association (APA) ethical standards for the treatment of research participants.

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Abstract

The study of infant cognitive and language development has been limited by small samples and the need for in-person visits. The Remote Infant Studies of Early Learning (RISE) (Tenenbaum et al., 2025) was designed to assess early development on a larger scale. The battery includes seven tasks to remotely assess developmental trajectories. This study expands on previous pilot work with a larger sample and modified methods. Data from 111 infants (55 6-month-olds and 56 12-month-olds) were collected asynchronously. Infant looking behavior was recorded with open-source software. Results for attention, memory, and prediction replicated expected patterns, supporting the feasibility of remote assessment in those domains. Other tasks assessing word comprehension and multimodal processing were not consistent with predictions, highlighting areas for further modifications of the remote tools. Overall, we demonstrate the translation and remote replication of several key tasks that assess early development which provides the opportunity for scalability of these tasks for larger, diverse populations.

Keywords: cognitive development, remote, infant

Introduction

Our understanding of early cognitive and linguistic development draws on more decades of laboratory-based studies that have generally required caregivers to travel to the lab with their infants for in-person assessments. While this approach brought us much of our foundational knowledge about infant cognitive development, a number of important findings have failed to replicate from one lab to the next (Lavelle, 2024). It is unclear whether these replication failures are because the findings are unreliable, the populations studied in the different locations differ in some way that has a meaningful effect on the results, or because the labs used different methods. Recent movements in "Big Science" (Byers-Heinlein, Tsui, Bergmann, et al., 2021; Byers-Heinlein, Tsui, van Renswoude, et al., 2021; Frank et al., 2017; Soderstrom et al., 2025; The ManyBabies Consortium, 2020; Zettersten et al., 2024) have improved our efforts to reproduce studies in larger samples, in some cases, with mixed results as some recent studies have failed to replicate seminal studies (Lucca et al., 2025; Schreiner et al., 2024).

The Remote Infant Studies of Early Learning (RISE) Battery was established to expand on these efforts using a single platform to present a consistent set of stimuli and one automated approach to responses. One of its overall aims is to facilitate the study of cognitive and linguistic development in early infancy in a scalable, generalizable manner; this in turn would allow for its application in other populations, such as infants with high risk of Autism Spectrum Disorders in future work. The RISE Battery consists of seven tasks modeled on traditional laboratory measures, adapted for remote use in the home. In conjunction with remote administration options through the Children Helping Science platform (Scott, Chu, & Schulz, 2017; Scott & Schulz, 2017) and automated coding of infant looking behavior via the open source software, iCatcher+ (Erel et al., 2023), the RISE Battery allows us to test developmental milestones without sources of unintentional experimenter bias and inconsistent stimuli presentation associated with lab-specific methods. Therefore, the RISE battery greatly reduces possible epistemic uncertainty. Recent work demonstrated the feasibility of this approach in a small sample of infants at 6 (n = 29) and 12 months (n = 26) (Tenenbaum et al., 2025). In the current study, we expand on those findings with (1) updated stimuli adapted from the pilot work and (2) a larger sample of infants.

The RISE Battery

The RISE Battery includes assessments of attention, memory, prediction, multisensory processing, word comprehension, social evaluation, and numeracy. These originally lab-based tasks were selected due to their demonstrated or hypothesized associations with developmental outcomes, including language skills, mathematical abilities, social skills, or autism diagnosis (Arunachalam & Luyster, 2016; Bradshaw, Shic, & Chawarska, 2011; Decarli, Piazza, & Izard, 2023; Hamlin, Wynn, & Bloom, 2007; Li et al., 2024; Pierce et al., 2016; Reuter, Emberson, Romberg, & Lew-Williams, 2018; Senju, 2013; Sinha et al., 2014; Starr, Libertus, & Brannon, 2013) (see Figure 1). With mixed success, the pilot study (Tenenbaum et al., 2025) added to growing evidence that established laboratory-based measures of language and cognitive development can be translated to in-home remote assessment (Bacon, Weaver, & Saffran, 2021; Bánki, de Eccher, Falschlehner, Hoehl, & Markova, 2022; Chuey, Boyce, Cao, & Frank, 2024; Silver et al., 2021).

In addition to establishing consistent methods for presentation of the experiment on the Children Helping Science platform, creating unified stimuli across infants, and exploring automated (i.e., scalable) coding methods, the RISE Battery adds a number of methodological advances that increase accessibility. These advances include testing infants in the comfort of their own homes, thereby reducing the demands and disruption of travel to a novel location and interaction with unfamiliar (and perhaps overwhelming) new experimenters. This approach also increases the feasibility of testing large populations (e.g., for population effect studies of environmental variables such as food or income supplements on infant development; in studies comparing populations across several locations, etc.). Further, it improves access for infants from special populations, including infant siblings of autistic children and infants with rare genetic profiles who may be too dispersed geographically to allow for in-person testing.

Though the described merits of a scalable automated assessment of infant cognitive and language development are clear, there are, of course, potential challenges associated with this approach as well. These include concerns about (1) whether we can recruit diverse participants given the requirement of a home computer, (2) whether infants will reliably attend to a full battery of experimental stimuli, and (3) whether data quality will be sufficient given potential variability in the internet connection speeds and possible dropouts in connection during the experiment. The current study assessed these concerns in 6 and 12-month-old infants recruited through the Children Helping Science Platform. In addition, we were interested in making modifications to the task battery used in Tenenbaum et al. (2025) in light of the data reported there, as well as adding new exploratory tasks to expand our coverage of different cognitive constructs. Below, we enumerate the constructs we measured and note modifications from the previous versions of these tasks, as relevant (see Figure 1 and Table 2).

Figure 1. Frame captures from the five base tasks (a) Geo-Social, (b) Visual Paired Comparison, (c) Prediction, (d) Multimodal Processing, (e) Word Comprehension, and two exploratory tasks (f) Helper-Hinderer, and (g) Numeracy

Attention: To explore social preference, infants viewed dynamic geometric shapes contrasted with children dancing (Chawarska, Ye, Shic, & Chen, 2016; Pierce et al., 2016). In our pilot work, 12-month-olds, but not 6-month-olds, looked reliably longer at the social stimuli (cf. (Imafuku et al., 2017)). Here, the stimuli were modified for younger participants by muting the colors of the geometric figures and zooming in on the faces of the children.

Memory: Visual paired comparison (VPC) (Manns, Stark, & Squire, 2000) was used to evaluate memory of both face and non-face stimuli ("Fribbles," which maintain the complexity of face processing while removing the social elements of the task (Barry, Griffith, De Rossi, & Hermans, 2014)). In the pilot study, results showed consistent preference for the novel face at test among 12-month-olds, but not 6-month-olds. Given the expected developmental gains between 6 and 12 months, this task was unchanged from the pilot study.

Prediction: The RISE Prediction task explores anticipatory looks to a cued visual location following familiarization. Previous research has shown correlations between this type of prediction and language abilities in infants without autism (Reuter et al., 2018). In the pilot study, infants consistently made anticipatory eye movements at 12 months but not at 6 months. Given the expected developmental shift, this task was unchanged from the pilot study.

Multimodal Processing: Language abilities have been associated with sensitivity to audiovisual alignment at young ages in both autistic and typically developing toddlers (Righi et al., 2018; Tenenbaum, Sobel, Sheinkopf, Malle, & Morgan, 2015). In the pilot study, infants did not show a significant preference for audiovisual (AV) synchrony when the audio preceded the video by 666ms. In the current study, we extended the delay to 1000ms.

Word Comprehension: To assess real-time word comprehension and its links to later language outcomes (Fernald & Marchman, 2012), infants view a looking-while-listening task in which they viewed two images, one of which is labeled (Bergelson & Swingley, 2012). Contrary to preregistered predictions, neither 6-month-olds nor 12-month-olds demonstrated reliable comprehension of these familiar words in the pilot study. Here, stimuli were adapted to be more similar to those used by Bergelson & Swingley (2012), by adding attention getters and reverting to previously tested stimulus pairs and images.

Social Evaluation: To evaluate prosocial evaluation, the RISE Battery uses animated stimuli in which social agents either push each other up (prosocial) or down (antisocial) a hill (Margoni & Surian, 2018). Though the pilot remote study did not replicate the previous findings, this result may have been related to the small sample size or translation to the remote setting. In the current study, the same pilot stimuli were used to assess social evaluation in infants in a larger sample.

Numeracy: In the pilot study, infants' non-symbolic number discrimination abilities were examined with a preferential-looking paradigm contrasting constant and alternating quantities of dots. Results suggested the ability to discriminate quantities differing by a 1:4-ratio (i.e., 5 vs 20 dots) at 12 months but not 6 months. Because the results among 12-month-olds were consistent with preregistered hypotheses, and the lack of effect among 6-month-olds may have been related to the small sample size in the pilot study, in the current study, the same pilot stimuli were used.

The hypotheses for the current study were the same as those set out in the pre-registration for the pilot study.

Methods

Ethical Compliance. The research presented in this manuscript was completed in compliance with the APA ethical principles regarding research with human participants. All methods were approved by the University Institutional Review Boards. The data reported here are part of an ongoing study in which infants with and without an older autistic sibling are tested on the RISE Battery, and caregivers are asked to complete surveys regarding their infant's developmental milestones. The current manuscript focuses on confirming the validity of the adapted RISE battery in a larger group of infants from the general population (i.e., without an autistic sibling).

Participants. Recruitment was completed through the Children Helping Science platform. Inclusion criteria were (1) >50% English spoken in the home, (2) full-term birth (>37 weeks), (3) without known developmental delays, and (4) without known relatives with a diagnosis of autism. A total of 111 participants were included with data collected at 6 months of age (6:0-6:30; n = 55) or 12 months of age (12:0-12:30; 56). The final sample included participants from 31 of the 50 states and was largely consistent with the racial composition of the United States (Scott et al., 2017), but participants were skewed towards higher education and income (see Table 1). Participants were compensated with a $25 Amazon gift card.

Stimuli. Except as noted, stimuli were consistent with those used in the pilot study (Tenenbaum et al., 2025). See Table 2.

Procedure. The Children Helping Science platform (formerly LookIt; (Scott et al., 2017; Scott & Schulz, 2017)) allows families to enroll in studies from their home computer. Consent was obtained through video recordings that were reviewed by study personnel before video data could be accessed. Families participated at their convenience and were encouraged to take breaks between tasks to promote attention to the stimuli. As long as the browser remained open, they were able to continue the experiment. The time between tasks ranged from 3 seconds to 73 minutes with a median of 11 seconds.

Video Coding. Infant gaze direction was detected using iCatcher+ (Erel et al., 2023), open-access software. iCatcher+ uses an artificial neural network to identify and classify gaze direction from low-resolution videos. To address concerns in the pilot study that the caregiver gaze was at times mistakenly identified as infant gaze (despite the fact

Table 1

Caregiver Demographics

Characteristic	Overall N = 111	Age 6 N = 55	Age 12 N = 56
Sex			
F	54 (48.6)	31 (56.4)	23 (41.1)
M	56 (50.5)	23 (41.8)	33 (58.9)
O	1 (0.9)	1 (1.8)	0 (0.0)
Race			
American Indian or Alaska Native	0 (0.0)	0 (0.0)	0 (0.0)
Asian	9 (8.3)	5 (9.4)	4 (7.1)
Black or African American	2 (1.8)	0 (0.0)	2 (3.6)
Native Hawaiian or Other Pacific Islander	0 (0.0)	0 (0.0)	0 (0.0)
White	57 (52.3)	28 (52.8)	29 (51.8)
More than one race	41 (37.6)	20 (37.7)	21 (37.5)
Ethnicity			
Hispanic	16 (14.7)	8 (15.1)	8 (14.3)
Not Hisp	93 (85.3)	45 (84.9)	48 (85.7)
Education			
High school	1 (0.9)	0 (0.0)	1 (1.8)
Associate (some college)	3 (2.7)	2 (3.7)	1 (1.8)
Bachelor/College	36 (32.7)	17 (31.5)	19 (33.9)
Graduate and Professional	70 (63.6)	35 (64.8)	35 (62.5)
Annual Income			
$<$ 50,000	11 (11.0)	6 (12.8)	5 (9.4)
50,000 - 100,000	41 (41.0)	19 (40.4)	22 (41.5)
101,000 - 200,000	25 (25.0)	13 (27.7)	12 (22.6)
$>$ 200,000	23 (23.0)	9 (19.1)	14 (26.4)
Geographic regions			
Northeast	31 (28.4)	17 (32.1)	14 (25.0)
Midwest	34 (31.2)	14 (26.4)	20 (35.7)
South	26 (23.9)	10 (18.9)	16 (28.6)
West	18 (16.5)	12 (22.6)	6 (10.7)

Note. Values are n (%). Percentages are calculated excluding missing data.

that iCatcher+ is designed to select the lowest bounding face), caregivers were asked to avert their gaze by looking down, away, or by wearing darkened glasses.

Table 2

The RISE Battery summary of results from the Pilot Study and modifications for the current study.

Task	Domain	Pilot Description	Pilot Results	Modifications
Geo-Social Attention	Attention	Preferential looking task to children dancing versus abstract geometric figures.	Preference for social stimuli in 12 but not 6-month-olds.	Zoomed in on faces and toned-down colors in geometric figures.
Visual Paired Comparison	Memory	Visual paired comparison for faces and abstract geometric figures.	Preference for novel face and object in 12 but not 6-month-olds.	None
Prediction	Pattern Recognition	Cued stimulus appears on one side of the screen $\times 8$ then on the opposite side $\times 8$.	Anticipatory eye movements in Block 2 for 12 but not 6-month-olds.	None
Audiovisual Synchrony	Multimodal Processing	Two social or nonsocial videos are shown side-by-side. One matches the audio track, the other precedes it by 666 ms.	Null results.	Increased delay to 1000 ms. Dropped non-social stimuli.
Word Comprehension	Language	Two images of highly familiar objects appear while infants hear ’Look at the [target]!’	Superior performance at 12 months, but no consistent increase in looks to the target in either age group.	Added attention getters. Reverted to previously tested stimulus pairs and images.
Social Evaluation	Social Congition	Animated shapes are shown to help or hinder each other’s progress up a hill.	Null results.	None
Numeracy	Number Representation	Alternating streams of different quantities of dots.	Preference for alternating at 12 but not 6 months.	None

Data quality & usable data. Frames were defined as usable if (1) the infant’s face was visible, (2) the infant’s gaze was directed at the screen, and (3) iCatcher+ classification confidence was .85 (see (Tenenbaum et al., 2025)); frames failing any criterion were excluded (exclusion counts are detailed below). Data quality metrics were evaluated across all administered trials for each task (see Table 3); note that the number of

Table 3

Data quality metrics from iCatcher+ output across all administered trials.

Task	N trials (QC)	N valid iCatcher+ (6)	N valid iCatcher+ (12)	N 0% completion (6)	N 0% completion (12)	Median (IQR) % no-face frames (6)	Median (IQR) % no-face frames (12)	Median (IQR) % away frames (6)	Median (IQR) % away frames (12)	Median (IQR) % low-confidence frames (6)	Median (IQR) % low-confidence frames (12)
Geo-Social Attention	10	54 (98.2%)	53 (94.6%)	0 (0.0%)	1 (1.9%)	0.0 (0.0, 2.7)	0.0 (0.0, 2.1)	5.3 (1.1, 11.4)	1.0 (0.0, 4.1)	9.9 (5.3, 14.6)	11.3 (7.7, 16.2)
Visual Paired Comparison (VPC)	12	50 (90.9%)	53 (94.6%)	2 (4.0%)	3 (5.7%)	5.5 (0.9, 18.1)	6.9 (1.7, 17.6)	21.9 (13.5, 28.2)	16.7 (10.5, 26.8)	9.8 (6.4, 14.6)	12.4 (9.3, 14.2)
Prediction	16	52 (94.5%)	51 (91.1%)	0 (0.0%)	0 (0.0%)	0.0 (0.0, 3.4)	0.0 (0.0, 5.8)	9.6 (1.7, 21.6)	5.8 (1.0, 13.5)	18.0 (10.8, 26.0)	22.1 (9.1, 35.1)
Audiovisual Synchrony	6	50 (90.9%)	54 (96.4%)	0 (0.0%)	0 (0.0%)	2.2 (0.3, 18.0)	3.0 (0.1, 8.5)	11.9 (7.5, 20.3)	9.5 (4.7, 22.5)	10.4 (6.5, 15.0)	11.7 (9.5, 18.1)
Word Comprehension	20	53 (96.4%)	53 (94.6%)	0 (0.0%)	0 (0.0%)	1.4 (0.0, 6.3)	0.9 (0.0, 7.8)	14.1 (9.4, 23.9)	7.5 (1.7, 13.9)	11.2 (7.4, 15.3)	13.3 (10.0, 18.7)
Numeracy	4	49 (89.1%)	52 (92.9%)	0 (0.0%)	0 (0.0%)	7.6 (0.8, 24.0)	6.4 (1.7, 18.7)	16.1 (4.3, 31.1)	19.5 (12.8, 34.6)	10.3 (6.6, 15.4)	15.3 (10.8, 17.3)
Social Evaluation	14	50 (90.9%)	48 (85.7%)	0 (0.0%)	0 (0.0%)	13.5 (4.9, 23.5)	6.8 (1.4, 14.0)	22.9 (12.3, 37.0)	14.4 (8.8, 24.1)	10.3 (7.9, 13.6)	13.5 (10.6, 15.8)

trials entering the analytic dataset differs from the total administered trials due to task-specific inclusion criteria described below (see Table 4 for analytic trial counts).

For the analytic dataset, for a trial to be considered usable, infants were required to have 33% usable frames. Participants were additionally excluded from a given task if they showed a side bias, defined as fixating on a single side for 80% of the task. Task-specific inclusion criteria were applied as follows. For the *Visual Paired Comparison (VPC) task*, each object was presented twice in succession (trials 1a and 1b); looking time was averaged across the pair, yielding 6 analytic units from 12 administered trials. For the *Prediction task*, trials 1 and 9 were excluded as the first trial of each block; participants were additionally required to have at least one anticipatory eye movement in Block 2 and at least two usable trials per block for the corresponding block outcome to be included (consistent with (Reuter et al., 2018)). The *Word Comprehension task* presented five word–image pairs across two repetition blocks (trials 1–10 and trials 11–20). For each pair, the target word appeared as the fixation target in one block and as the distractor in the other (e.g., for the pair foot-milk, foot was the target in trial 1 and the distractor in trial 9, and vice versa.)

The outcome for each pair was the proportion of time infants looked at an item when it was the target (across both blocks of trials) minus how much they looked at that same item when it was the distractor (across both blocks of trials), as in Bergelson & Swingley, 2012. This yields 5 pair-level difference scores per participant. For the *Social Evaluation task*, only trials 7 and 14 served as test trials (one per block); participants were required to have at least one usable helper trial and one usable hinderer trial within each block (Block 1: trials 1–6; Block 2: trials 8–13) for the corresponding test trial to be included. No additional task-specific criteria were applied for Geo-Social Attention, Audiovisual Synchrony, or Numeracy. The number of analytic trials, therefore, differs from the total administered trials for four tasks; Table 4 summarizes analytic sample size and trial counts.

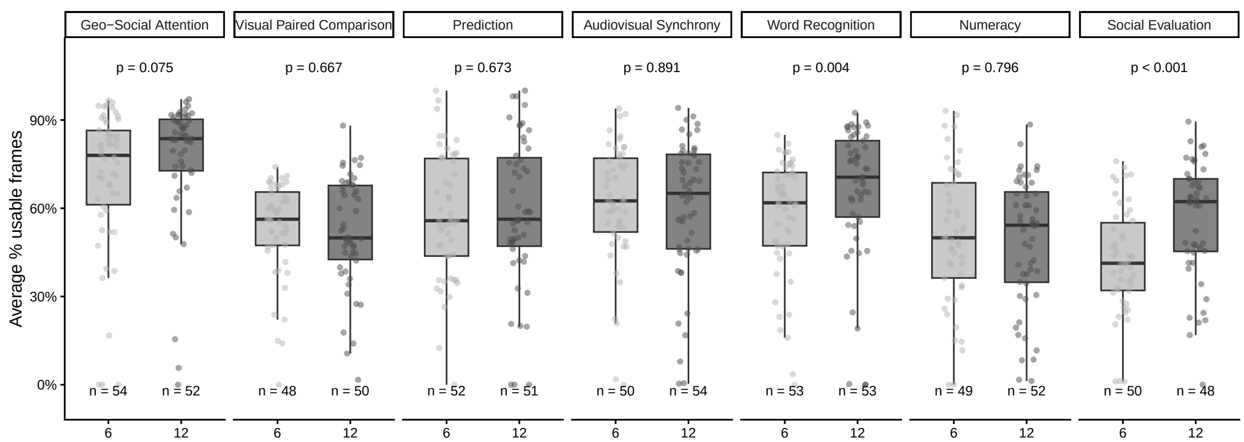

Figure 2. Average percentage of usable frames per participant by task across all administered trials.

Statistical Analysis. Demographics and participant characteristics were summarized using median and interquartile range (IQR) for continuous variables and frequency and percentage for categorical variables. The dependent variable differed by task. For the preferential looking tasks (Geo-Social Attention, Visual Paired Comparison, Audiovisual Synchrony, Numeracy, and Social Evaluation), the dependent measure was the proportion of looks to the target stimulus relative to total looking time to the screen, treated as continuous with an expected proportion greater than 0.5. For the Word

Comprehension task, the dependent measure was a difference score calculated as the proportion of attention to the target image when it was named by the auditory stimulus minus the same proportion when it served as the distracter image, averaged across trials. This was treated as continuous with an expected mean difference score greater than 0. For the Prediction task, the dependent measure was the proportion of anticipatory eye movements (AEMs) to the target side, calculated as the number of usable trials with at least one AEM divided by the total number of usable trials, separately for each block (Block 1: trials 2–8; Block 2: trials 10–16), with an expected mean proportion greater than 0.5.

All statistical analyses were conducted in R (version 4.5.2) using the packages lme4 and lmerTest. Linear mixed-effects models were fit with child and stimulus as crossed random effects to test for expected patterns in the overall cohort and by age group (e.g., proportion_of_looks_to_target ~ age + (1|child_id) + (1|trial)). Three tasks required additional model terms to account for within-task design features. For the Visual Paired Comparison task, a Stimulus Type term (faces vs. fribbles) was included as a within-participant factor, as all infants viewed both stimulus types; an Age × Stimulus Type interaction term was included to test whether novelty preferences differed by stimulus type across age groups. For the Prediction task, a Block term (Block 1 vs. Block 2) was included as a within-participant factor, as all infants contributed data to both blocks; an Age × Block interaction was included to examine whether anticipatory eye movements differed across the two blocks. For the Numeracy task, Constant Stream Size (5-item vs. 20-item) was included as a between-participant factor, as infants were randomly assigned to view only one constant stream condition for the entire task; an Age × Constant Stream Size interaction was included to test whether preferences for the alternating stream differed by condition. The statistical analyses closely followed the preregistered analysis plan. To better reflect the within-participant design of several tasks, mixed-effects models for the Visual Paired Comparison, Prediction, and Numeracy tasks included interaction terms (Age × Stimulus Type, Age × Block, and Age × Constant Stream Size, respectively). With the Prediction task, we modeled Block as a within-participant factor in a single mixed-effects model rather than fitting separate models for overall performance and Block 2. Overall, with these modifications, we tested the preregistered hypotheses without changing the primary outcomes.

Where consent was provided, participant videos are available on Databrary. Code and data are available through GitHub.

Results

Infant Attention and Side Bias Across All Administered Trials. Table 3 summarizes data quality metrics across tasks, including the proportion of frames excluded due to no face detected, off-screen gaze, and low iCatcher+ confidence. Valid iCatcher+ output was obtained for the majority of participants across all tasks (range: 85.7%–98.2%). Zero percent task completion was rare, with the exception of the VPC task where 4.0% of 6-month-olds and 5.7% of 12-month-olds had no usable trials. Across tasks, frame exclusions were driven by all three sources, though away looks were consistently the largest contributor relative to no-face frames. Low-confidence frames were also notably prevalent, particularly for the Prediction task (18.0% and 22.1% for 6- and 12-month-olds, respectively). Social Evaluation showed the highest rates of both no-face and away frames (no-face: 13.5% and 6.8%; away: 22.9% and 14.4% for 6- and 12-month-olds, respectively).

Figure 2 displays the average percentage of usable frames per participant across the seven tasks. Overall, usable frame rates were moderate to high, with median values ranging from approximately 50% (Social Evaluation, 6-month-olds) to 85% (Geo-Social Attention, 12-month-olds), suggesting adequate data quality for most participants. Wilcoxon rank-sum tests indicated that 12-month-olds had significantly higher usable frame rates than 6-month-olds for Word Comprehension (p = .018) and Social Evaluation (p < .001);

no significant age differences were observed for the remaining five tasks. There was no significant difference in usable frame rates as a function of task presentation order for either age group across the five base tasks (all ps > .05; see Supplementary; Figure 1 for tasks separation) or the two exploratory tasks, Social Evaluation and Numeracy.

Table 4

Participant retention across data quality and inclusion criteria steps, by task and age group.

Task	Analytic trials/word pairs	N valid iCatcher+ (6)	N valid iCatcher+ (12)	N 33% usable frames (6)	N 33% usable frames (12)	N side bias (6)	N side bias (12)	N final analytic sample (6)	N final analytic sample (12)	Median (IQR) trials (6)	Median (IQR) trials (12)
Geo-Social Attention	10	54 (98.2%)	53 (94.6%)	51 (94.4%)	50 (94.3%)	9 (17.6%)	3 (6.0%)	42	47	8.0 (7.0, 9.8)	8.0 (7.0, 9.0)
Visual Paired Comparison	6	50 (90.9%)	53 (94.6%)	47 (94.0%)	47 (88.7%)	6 (12.8%)	5 (10.6%)	41	42	6.0 (4.0, 6.0)	5.0 (4.0, 6.0)
Prediction	14	52 (94.5%)	51 (91.1%)	51 (98.1%)	48 (94.1%)	—	—	37	35	10.0 (7.0, 12.0)	10.0 (8.0, 12.5)
Audiovisual Synchrony	6	50 (90.9%)	54 (96.4%)	47 (94.0%)	50 (92.6%)	11 (23.4%)	10 (20.0%)	36	40	6.0 (5.0, 6.0)	6.0 (5.0, 6.0)
Word Comprehension	5	53 (96.4%)	53 (94.6%)	52 (98.1%)	50 (94.3%)	13 (25.0%)	11 (22.0%)	37	38	5.0 (4.0, 5.0)	5.0 (4.2, 5.0)
Numeracy	4	49 (89.1%)	52 (92.9%)	45 (91.8%)	43 (82.7%)	18 (40.0%)	12 (27.9%)	27	31	4.0 (3.0, 4.0)	3.0 (2.5, 4.0)
Social Evaluation	2	50 (90.9%)	48 (85.7%)	26 (52.0%)	37 (77.1%)	3 (11.5%)	7 (18.9%)	23	30	2.0 (1.0, 2.0)	2.0 (1.0, 2.0)

Analytic Dataset. Table 4 summarizes participant retention across inclusion criteria steps. Attrition was most pronounced for Social Evaluation, where only 52.0% of 6-month-olds with valid iCatcher+ output met the usable trial threshold, and for Numeracy, where side bias exclusions resulted in the smallest final analytic samples (6-month-olds: n = 27; 12-month-olds: n = 31). Participants who fixated on a single side of the screen for 80% of usable frames within a task were excluded from that task's analysis; this criterion was not applied to the Prediction task due to its short anticipatory eye movement analysis window. Note that for Prediction, Word Comprehension, and Social Evaluation, the final analytic N reflects additional task-specific criteria beyond side bias exclusion (see Methods); these are reflected in Table 4.

Task Outcomes. Results are reported as model-based estimates, corresponding 95% confidence limits, and p-value of the corresponding hypothesis test (with significance defined as two-sided $p < 0.05$).

Geo-Social Attention Task. The pre-registered prediction for the Geo-Social Attention Task was that infants would look more (mean > 0.50) to the social stimuli. This prediction held true for both 12- and 6-month-olds (Figure 3a) 12-m: M = 0.70 [0.64, 0.76], $p = < 0.001$; 6-m: M = 0.66 [0.60, 0.72], $p = < 0.001$) without a significant effect of age (0.04 [-0.01, 0.09], $p = 0.127$). These findings are consistent with the results of the 12-month-olds in the pilot study and with prior work in toddlers Pierce et al., 2011; Tenenbaum et al., 2025), and further show that this preference is present in infants as young as 6 months (as in Imafuku et al., 2017). See Figure 3a and Table 5.

Visual Paired Comparison (VPC) Task. Our preregistered hypothesis was that 12-month-olds would show a stronger novelty effect than 6-month-olds and that the effects of novelty would be stronger for faces than for Fribbles. The results showed that 12-month-olds but not 6-month-olds demonstrated significant preference for novel faces (12-m: M = 0.58 [0.51, 0.66], $p = 0.033$; 6-m: M = 0.56 [0.49, 0.64], $p = 0.084$), without a significant effect of age ($p = 0.508$). For non-face objects (Fribbles), 6-month-olds (M = 0.56 [0.51, 0.62], $p = 0.026$), and 12-month-olds (M = 0.56 [0.51, 0.62], $p = 0.025$) showed a reliable preference for the novel object at test, and again there was not a significant effect of age ($p = 0.969$). The effect of stimulus type was not significant for either 6- or 12-month-olds ($p = 0.98$ and $p = 0.62$, respectively). See Figure 3b and Table 5.

Prediction Task. The pre-registered hypothesis was that 12-month-old infants would demonstrate more AEMs to the target side (versus the distractor side) than 6-month-old infants. As seen in Table 5, there were more AEMs to target amongst the 12-month-olds; however, these differences between age groups were not significant (Block 1: 12-m: M = 0.52 [0.39, 0.64], $p = 0.778$; 6-m: M = 0.38 [0.26, 0.50], $p = 0.044$; 12-m vs. 6-m: M = -0.14 [-0.31, 0.03], $p = 0.108$. Block 2: 12-m: M = 0.68 [0.55, 0.80], $p =$

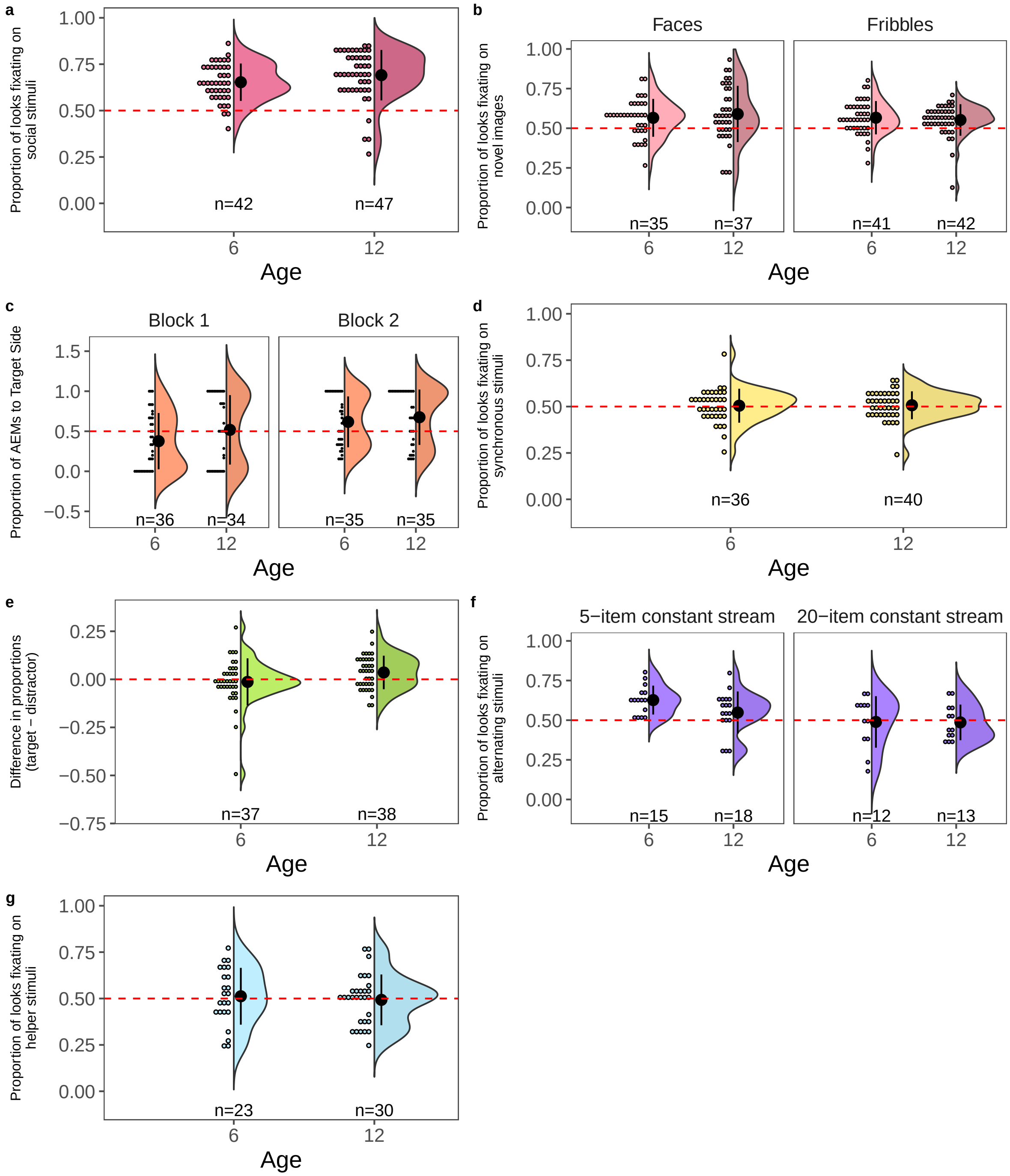

Figure 3. Results for (a) Geo-Social Attention, (b) Visual Paired Comparison, (c) Prediction, (d) Audiovisual Synchrony, (e) Word Comprehension, (f) Numeracy, and (g) Social Evaluation.

0.005; 6-m: M = 0.62 [0.50, 0.74], $p = 0.059$; 12-m vs. 6-m: M = -0.06 [-0.23, 0.11], $p =$ 0.507). As reflected in these analyses, both 6-month-olds and 12-month-olds failed to show reliable AEMs to the target side in Block 1. In Block 1, 6-month-olds overall were even significantly not showing anticipatory looks towards the target side but rather looking towards the incorrect side. The pattern with the 12-month-olds was also surprising given that the 12-month-olds seem to show reliable predictions to the updated location in Block 2. (See Figure 3c and Table 5.)

To clarify the source of these unexpected results, we ran an exploratory analysis comparing the proportion of infants with AEMs to chance on each trial (uncorrected). In Block 1, results indicated that overall, neither 6-month-olds nor 12-month-olds reliably demonstrated AEMs on any individual trial (Figure 4). In Block 2, most 12-month-olds were able to update their predictions about the new location reliably by Trial 11 (the 2nd trial in Block 2), whereas 6-month-olds were able to do so on Trial 11 and Trial 14 but were not consistent on other trials (and as such on average across the trials in Block 2 did not show a reliable tendency to predict the updated location).

Audiovisual Synchrony Task. In contrast to our pre-registered predictions, we did not find a reliable preference for the synchronous stimuli at either age (12-m: M = 0.51 [0.46, 0.55], $p = 0.742$; 6-m: M = 0.50 [0.46, 0.55], $p = 0.851$), nor was there a significant effect of age ($p = 0.897$). See Figure 3d and Table 5.

Word Comprehension Task. Again, in contrast from our preregistered hypothesis, the 12-month-olds did not differ from the 6-month-olds in their looks to the target (M = 0.04 [0.00, 0.08], $p = 0.042$). Neither the 12-month-olds nor the 6-month-olds looked more to the target object when it was the target than when it was the distractor (12-m: M = 0.04 [-0.00, 0.08], $p = 0.077$; 6-m: M = -0.01 [-0.05, 0.04], $p = 0.773$). See Figure 3e and Table 5.

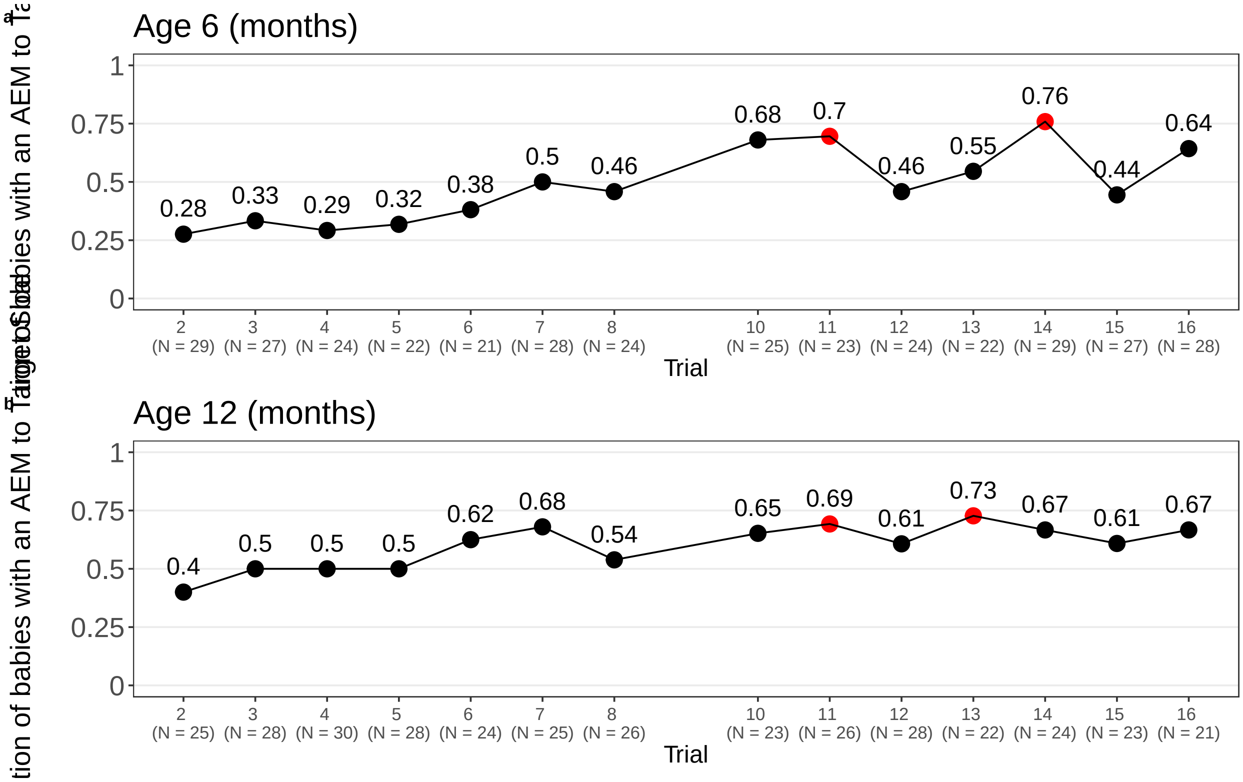

Figure 4. Proportion of infants showing anticipatory eye movements (AEMs) to the target side by trial, for (a) 6-month-olds and (b) 12-month-olds (Block 1: Trials 2–8; Block 2: Trials 10–16).

Numeracy Task. In the Numeracy task, we predicted that infants at both 6 and 12 months would prefer the numerically alternating streams compared to the constant streams irrespective of the relative size of the quantity in the constant stream (i.e., 5 or 20 items). Contrary to our predictions, 6-month-olds preferred, and 12-month-olds trended toward preferring, the alternating stream only when the constant stream contained the smaller quantity (12-m: M = 0.56 [0.49, 0.63], p = 0.082; 6-m: M = 0.63 [0.56, 0.70], p = 0.001) not when the constant stream contained the larger quantity (12-m: M = 0.49 [0.41, 0.57], p = 0.864; 6-m: M = 0.49 [0.40, 0.57], p = 0.738). There was no significant effect of age for large or small constant (p = 0.902 and p = 0.147, respectively). See Figure 3f and Table 5.

Social Evaluation Task. Inconsistent with preregistered predictions, we found no reliable difference in looks to the helper character relative to the hindering character (12-mo: M = 0.49 [0.43, 0.54], $p = 0.657$; 6-mo: M = 0.51 [0.45, 0.57], $p = 0.665$) and no significant effect of age ($p = 0.537$). See Figure 3g and Table 5.

Table 5

Model-estimated mean proportion looking time to target by task and age group, with age group comparisons.

Task	Estimate (95% CI)	p	6 months Estimate (95% CI)	6 months p	12 months Estimate (95% CI)	12 months p	Age Effect Estimate (95% CI)	Age Effect p
Geo-Social Attention	0.68 (0.63, 0.73)	< 0.001	0.66 (0.60, 0.72)	< 0.001	0.70 (0.64, 0.76)	< 0.001	0.04 (-0.01, 0.09)	0.127
VPC Social (face)	0.57 (0.50, 0.65)	0.053	0.56 (0.49, 0.64)	0.084	0.58 (0.51, 0.66)	0.033	-0.02 (-0.08, 0.04)	0.508
VPC Non-social (fribbles)	0.56 (0.51, 0.62)	0.029	0.56 (0.51, 0.62)	0.026	0.56 (0.51, 0.62)	0.025	-0.00 (-0.04, 0.04)	0.969
Prediction Block 1	0.45 (0.36, 0.53)	0.227	0.38 (0.26, 0.50)	0.044	0.52 (0.39, 0.64)	0.778	-0.14 (-0.31, 0.03)	0.108
Prediction Block 2	0.65 (0.56, 0.73)	0.001	0.62 (0.50, 0.74)	0.059	0.68 (0.55, 0.80)	0.005	-0.06 (-0.23, 0.11)	0.507
AV Synchrony	0.51 (0.46, 0.55)	0.765	0.50 (0.46, 0.55)	0.851	0.51 (0.46, 0.55)	0.742	0.00 (-0.04, 0.05)	0.897
Word Recognition	0.02 (-0.03, 0.06)	0.36	-0.01 (-0.05, 0.04)	0.773	0.04 (-0.00, 0.08)	0.077	0.04 (0.00, 0.08)	0.042
Numeracy 5-item stream constant	0.59 (0.54, 0.65)	0.007	0.63 (0.56, 0.70)	0.001	0.56 (0.49, 0.63)	0.082	0.07 (-0.02, 0.16)	0.147
Numeracy 20-item stream constant	0.49 (0.43, 0.55)	0.728	0.49 (0.40, 0.57)	0.738	0.49 (0.41, 0.57)	0.864	-0.01 (-0.12, 0.10)	0.902
Social Evaluation	0.50 (0.46, 0.54)	0.98	0.51 (0.45, 0.57)	0.665	0.49 (0.43, 0.54)	0.657	-0.03 (-0.11, 0.06)	0.537

Summary in Relation to Pilot Study. In summary, the current study replicated pilot results with Geo-Social (12-month-olds), VPC and Prediction tasks (12-month-olds). Changes to the stimuli (and overall increased data collection) as described in Table 2 resulted in outcomes consistent with the prior literature for 6-month-olds in the Geo-Social task. Audiovisual Synchrony, Social Evaluation, and Word Comprehension failed to show the predicted outcomes. The larger sample size shifted the interpretation of the Numeracy findings to be associated with increased numbers of items on the screen, rather than solely discrimination of quantities, per se.

Discussion

The present study provides more evidence towards using the Remote Infant Studies of Early Learning battery to assess multiple mechanisms that serve as the building blocks for early development through a remote, at-home accessible platform. This not only serves to further understand and assess both typical and atypical development. The RISE Battery allows for unbiased assessment via presentation of a standardized set of stimuli via one platform, with video coding of infant behaviors using a single, consistent, automated process. This approach increases infant comfort during assessment by allowing the families to participate from home without interaction with novel settings and people in the context of their engagement with the stimuli. This approach also displays a continuing effort to replicate studies over large samples. The initial pilot study (Tenenbaum et al., 2025) found predicted results for a subset of the tasks (Geo-Social, Visual Paired Comparison, Prediction, and Numeracy) in a group of infants recruited from the general population. Here, we demonstrate replication (Geo-Social, Visual Paired Comparison, Prediction) of those results from our pilot study.

Task-Based Results. Consistent with decades of research showing that young infants are drawn to face stimuli (Fantz, 1964; Johnson, Dziurawiec, Ellis, & Morton, 1991), both 12- and 6-month-olds showed a preference for social stimuli in the Social

Attention Task. The current results extend our pilot findings down to 6-month-olds and could provide a means to test for social preference in infants with an elevated likelihood of atypical social development.

In the Visual Paired Comparison Task, we found that both 6- and 12-month-old infants showed a preference for novel stimuli overall (both faces and objects). We did not find significant effects of age or stimulus type.

In the Prediction Task, we measured prediction to a cued visual stimulus over a single trained location and then an alternate location that required updated predictions. Within-trial results suggest that in Block 1, both 12-month-olds and 6-month-olds did not reliably predict the new location on any trial. However, in Block 2, 12-month-olds recognized the pattern by the eleventh trial, whereas 6-month-olds did do so in the eleventh trial but not reliably throughout Block 2. This capacity to update one's predictions in the visual domain was previously linked to language abilities (Reuter et al., 2018). Data collection for language outcomes (via a vocabulary checklist) to address this question in the current sample is underway.

In the Audiovisual Synchrony task, we did not find the anticipated preference for the synchronous stimuli at either age. From the pilot study, we increased the asynchronous delay between the audio and video from 666ms to 1000ms. however, this increase was not effective in prompting and observing a preference for the synchronous stimuli. While previous work identified a reliable preference for synchronous videos in typically developing toddlers (M = 3.04 years; range: 1-7 years) (Righi et al., 2018), the lack of effect here is likely due to the younger age of the participants. Few studies have observed this preference with younger infants, with one recent work observing this pattern as early as 18-months (Clark et al., 2026). It is also possible that remote administration of these multimodal stimuli was affected by variable internet connection reliability.

Between the pilot and current versions of the Word Comprehension task, we increased the number of attention getters from 1 (every 10 trials) to 3 (every 5 trials) and used the a subset of word pairs from Bergelson and Swingley (2012). Of note, as compared to the original Bergelson and Swingley (2012) study, the current task had substantially fewer trials, and trial structure diverged somewhat (in the original study, parents produced the sentences and images were onscreen for 3.5s after target word onset; here trials were somewhat shorter, with reduced critical windows. These differences, intended to reduce task length for the battery, may have contributed to the lack of success among the infants. Future versions of the task have been updated to maintain slightly longer trials to match the critical window of the original study.

For Numeracy, both 6- and 12-month-olds looked longer at the numerically alternating stream (i.e., the stream alternating between 5 and 20 dots), but only when the constant stream contained the smaller quantity (i.e., 5 dots). This suggests that the effect may have been driven in part by infants attending more to larger numbers of objects. The fact that infants did not significantly prefer the constant stream when it contained the larger quantity suggests that infants' preference may be driven by a mix of attention to the alternating quantities and the large quantity. Regardless, our findings do not support the claim that infants' preferences were solely reflective of discrimination of the alternating numbers of objects in a stream. Importantly, this version of the task was shortened relative to the studies on which it was based (from 60 to 20 seconds per trial) (Libertus & Brannon, 2010; Starr et al., 2013). Further, the current study differed from previous instantiations in that stimuli were presented on a single screen (rather than on two separate screens).

In the Social Evaluation task, our results were similar to those from the pilot study, with infants at 6 months and 12 months showing no preference for the helper. Previous lab research has shown that infants show a reliable preference for the prosocial agent (the helper) in this task when using a habituation paradigm in many individual lab studies, but this effect was not found in a large multi-lab study (see review in (Lucca et al., 2025; Margoni & Surian, 2018)). Our continued finding of no preference for the helper character and no age effects suggests a need to reframe the task. Future remote work may adjust the task by presenting side-by-side displays of helping versus hindering events to better capture infants' preferences with the remote preferential-looking paradigm.

Limitations

Though not outside the range of typical performance among infants, even in laboratory-based experiments (Chawarska, Macari, & Shic, 2013; Elsabbagh et al., 2009; Frank, Vul, & Johnson, 2009; Narayan, Werker, & Beddor, 2010; Peltola, Leppänen, Mäki, & Hietanen, 2009), substantial data loss is a limitation of this battery. Additionally, this study requires access to adequate high-speed internet and a laptop, which may lead to overrepresentation of high-SES families and that could ultimately undermine some of the clinical generalizability of the findings. Though the use of remote methods was designed to improve the diversity of participant samples relative to laboratory-based studies (Scott et al., 2017; Scott & Schulz, 2017), we are still not reaching a sample that is representative of the US population with respect to SES. Given previously identified relationships between SES and early language, for example (Fernald, Marchman, & Weisleder, 2013; Huang, Byrd, Asmah, & Domanski, 2023), establishing predicted patterns of performance with predominantly high-SES populations creates risk for over-identification of divergence in developmental trajectories among lower-SES families.

Conclusions

This work provides a demonstration of the feasibility of remote assessment and automated coding of group-level measures of infant abilities in tasks assessing attention (Geo-Social task), memory (Visual Paired Comparison), learning (Prediction). The replication of those task exemplify the robustness of both the tasks and the use of the remote setting for those tasks. We plan to continue our investigation of the potential for this battery of tasks to inform our understanding of early cognitive development in large samples of infants so that we can better understand varied developmental trajectories. This approach has significant potential to allow for assessment of large populations of infants that could make it more feasible to assess the effects of interventions that require extremely large samples (e.g., food and income supplement programs). This approach also provides a way to quickly test large populations, asynchronously, at times that work best for families. Without remote methods, many populations are unintentionally excluded from testing. The RISE battery could improve the feasibility of assessing challenging-to-reach populations such as those at elevated likelihood for autism spectrum disorder due to the presence of a sibling with autism, and to those with rare genetic conditions. This approach highlights the potential to test such populations using the same automated measures. Finally, for interventions targeting child development a single mode of assessment that can be administered in multiple locations could allow us to examine how different interventions function in different communities around the world.

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